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Atrial fibrillation recurrence and spontaneous conversion to sinus rhythm after cardiac surgery: Insights from 426 patients with continuous rhythm monitoring

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Abstract

Introduction: Cardiac surgery may influence occurrence and persistence of atrial fibrillation (AF). In patients with a history of cardiac surgery and postoperative pacemaker or implantable cardioverter defibrillator implantation we attempted to quantify AF characteristics and investigate factors that influence AF recurrence.

Methods: In 426 patients (319 male; age: 73 *SD*: 9 years) with a history of cardiac surgery (364 CABG; 69 mitral valve; 57 aortic valve and 59 combination procedures) the heart rhythm history (mean: 351 days *SD*: 54 days) was reconstructed and investigated for AF recurrence patterns.

Results: AF developed and subsequently recurred in 75% of patients ($n = 320$). The mean AF burden in these patients was 0.21 (21% of the total observed time spent in AF) and 4085 episodes of AF were observed and analyzed. AF episodes spontaneously terminated within 24 h ($n = 2509$), 48 h ($n = 700$), 72 h ($n = 279$), and 1 week ($n = 31$). The probability of spontaneous conversion to sinus rhythm (SR) decreased with time spent in AF and plateaued after approximately 7 days. Patient age, LVEF and presence of coronary artery disease significantly influenced the probability of spontaneous conversion to SR. Type of cardiac surgery also significantly influenced AF episode characteristics.

Conclusion: AF episodes are common in patients with a history of cardiac surgery. Spontaneous conversion to SR diminishes with increasing time spent in AF and is influenced by AF characteristics and several patient-related factors. Continuous monitoring can provide detailed information about AF recurrence that may immensely improve our understanding and influence the clinical management of AF.

KEYWORDS

atrial fibrillation, atrial fibrillation recurrence, cardiac implantable electronic devices, cardiac surgery, continuous heart rhythm monitoring

1 | INTRODUCTION

The recurrence of atrial fibrillation (AF) is regarded a random phenomenon. Medical as well as interventional therapies can affect AF recurrence, by either reducing the amount of time a patient spends in AF (AF burden) or stabilizing the patient's electrophysiological substrate to avoid AF recurrence, or prolonging the time to the next AF recurrence.[1] Cardiac surgery has been shown to induce electrical instability and affect AF recurrence in patients with and without a previously known history of AF.[2]

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Continuous monitoring (CM) with implantable heart rhythm recorders[3] and even wearable electronics[4] has been shown to reliably detect AF. These devices have introduced a change of mentality from a binary consideration of AF (yes/no) to a quantitative concept of AF (how much AF: AF burden). We have previously shown that intermittent monitoring is vastly inferior to CM for AF detection,[5,6] AF quantification[7,8] and even AF classification.[9] Data from CM devices provide the opportunity to quantify AF recurrence,[5] identify patterns of AF recurrence and its temporal aggregation,[10] and provide a more quantitative AF classification.[9]

The aim of the present work was two-fold. First, we sought to quantify the AF recurrence characteristics in patients with a history of cardiac surgery. Second, based on this population subset, we sought to investigate if patient or AF characteristics influence the probabilities of spontaneous conversion to sinus rhythm (SR). To our knowledge the present study is the only largescale study evaluating AF recurrence in cardiac surgery patients using continuous rhythm monitoring.

2 | METHODS

2.1 | Patient population

The patient population included for this analysis is the subset of patients with a history of cardiac surgery enrolled in the OMNI[11] and TRENDS[12] studies. The complete patient population has been previously published in detail.[9] From the initial population of the OMNI ($n = 737$) and TRENDS ($n = 598$) trials and for the purposes of the present analysis, we excluded 60 patients with AF specific treatments (medical/electrical cardioversion or catheter ablation), 27 patients with single chamber devices, and 7 patients in whom no atrial lead was implanted. Participation in the trials means that these are all patients who had received pacemakers (PM) or implantable cardioverter defibrillators (ICD) at some point after surgery. The time between device implantation and surgery was a minimum of 3 months. As the present work is an analysis of data collected in the past for the purpose of the OMNI and TRENDS multicenter trials a renewed approval from an ethics committee for the purposes of the present study was not required. In the present analysis, we focus on the 426 patients with a history of cardiac surgery. No patient had an attempted cardioversion, AF ablation (surgical/interventional) or antiarrhythmic medical therapy anytime during follow-up, underwent device replacements, already had permanent atrial tachycardia/AF, had known reentrant supraventricular tachycardia, or had a terminal illness. All patients had at least 180 days of documented rhythm history from the device trending data (Cardiac Compass; Medtronic Inc.). All devices in this population had an atrial lead for the detection of atrial activity. The atrial activity detected by the devices constitutes the basis of the data evaluated regarding AF recurrence. Patient population characteristics are presented in Table 1.

TABLE 1 Demographic, device, and operative characteristics of the investigated cohort

		Nr./Mean	(%/*SD*)
Total cohort	TRENDS and OMNI	426	100.0
Source Studies	TRENDS	210	49.3
	OMNI	216	50.7
Source of monitoring data: Device type	CRT	113	26.5
	ICD	154	36.2
	PM	159	37.3
Clinical AF classification	Paroxysmal	389	91.3
	Persistent	37	8.7
Demographic data	Age (years)	72.76	9.0
	Male gender	319	74.9
	Follow-up (days)	351	54
	LVEF (%)	38	16
	Coronary artery disease	376	88.3
	Hypertension	319	74.9
	Diabetes	149	35.0
	Cardiomyopathy	268	62.9
	History of arrhythmias		
	Atrial flutter	78	18.3
	Atrial tachycardia	15	3.5
	NYHA functional class		
	None	113	26.5
	I	39	9.2
	II	111	26.1
	III	106	24.9
	IV	9	2.1
	unknown	11	2.6
Operative details	Type of cardiac surgery		
	CABG	364	85.4
	AVR	57	13.4
	MVR	69	16.2
	TVR	8	1.9
	Combined procedures		
	All combined procedures	59	13.8
	2	49	11.5
	3	7	1.6
	4	3	0.7

Abbreviations: AF, atrial fibrillation; AVR, surgical aortic valve reconstruction/replacement; CABG, coronary artery bypass graft surgery; CRT, cardiac resynchronization therapy; ICD, implantable cardioverter defibrillator; LVEF, left ventricular ejection fraction; MVR, mitral valve reconstruction/replacement; NYHA, New York Heart Association functional classification; PM, pacemaker; TVR, tricuspid valve reconstruction/replacement.

2.2 | AF burden and density calculations

In total, 4085 episodes of AF were observed and included in this analysis. Because during a period of arrhythmia and due to vulnerable AF substrate, patients may alternate between AF and SR in rapid succession indicating an electrically unstable substrate, we considered an episode of AF to be terminated only if it was succeeded by 24 h of stable, uninterrupted SR. The nominal detection setting for AF detection was an atrial rate greater than 175 bpm which lasted ≥20 s. Episode determination was made only by the device and each episode was not reviewed by an investigator. However, AF detection in these devices with an atrial lead has been shown to have high accuracy, particularly for episodes longer than 6 min in duration. Atrial flutter and atrial tachycardia were not differentiated from AF as long as the AF detection criteria were satisfied.

To identify and quantify patterns of AF recurrence, we calculated the AF burden as the proportion of observation time that a patient spends in AF. For example, a patient with an AF burden of 0.15 has spent 15% of the total observation time in AF. This alone does not reveal anything about the distribution of the AF duration. We have previously developed the concept of AF density.[5,10] The AF density takes values (0,1) and low AF density denotes AF recurrence that is more spread apart during the observation period while high AF density denotes AF episodes that are aggregated together. With Figure 1 we aim to underline the concept described above showing two patients with the same amount of AF recurrence (same AF burden) but different recurrence patterns (high density AF [Patient A] vs. low density AF [Patient B]).

2.3 | Calculating the probability of spontaneous conversion

For all 426 patients and for all 4085 AF episodes during the 409.6 patient*years of follow-up, we calculated the conditional probability of spontaneous conversion on day T = t given the patient has AF until day T as Pr(Conversion at T = t | AF episode duration ≥ t). For every patient i a logistic regression model m_i of the form: log(Pr(Conversion at T = t; patient i | AF episode duration ≥ t; patient i)/1-Pr(Conversion at T = t; patient i | AF episode duration ≥ t; patient i) = $b_{0i} + b_{1i}$ln (days in AF) was calculated, collecting for each patient the patient level estimates b_{0i} and b_{1i}. Goodness of fit test (Hosmer–Lemeshow test) as well as visual inspection of fitted and predicted probabilities was performed. Due to data acquisition limitations of the devices used, AF episode durations of ≤1 day have been aggregated.

After obtaining the patient level estimates b_{0i} and b_{1i} we then regressed these estimates with various AF-related and patient-related factors to identify predictors that influence the probability of conversion with time. These linear models were of the form b_0 = intercept + a × regressor; b_1 = intercept + a × regressor. AF characteristics (AF burden, defined as the proportion of observation time a patient spends in AF), temporal aggregation of AF (AF density), as well as patient-related variables were investigated: patient age, gender, left ventricular ejection fraction (LVEF), presence of coronary artery disease (CAD), AF classification (paroxysmal/persistent), history of hypertension, diabetes, NYHA functional class, history of cardiac surgery. When a statistically significant influence of a patient factor could be identified, we additionally reevaluated if the influence of this factor persisted when AF burden and AF density were forced in the model. After obtaining the factors that exhibit a significant influence on b_0 and b_1 for both conversion to SR and recurrence, we used stepwise bidirectional regression to build multivariable models for b_0 and b_1.

We report our results as the coefficients of the regressors on b_0 and b_1. One can then estimate the probability for conversion to SR or AF recurrence by the back transformation:

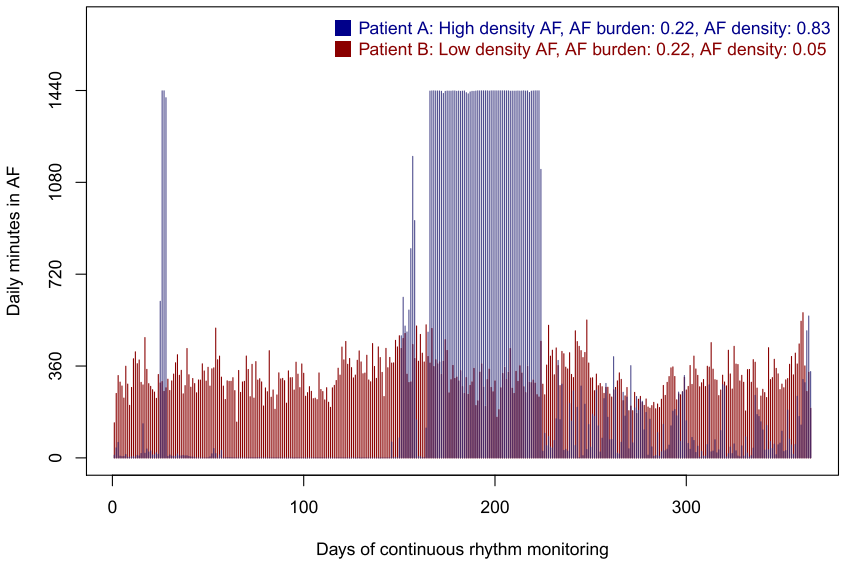


FIGURE 1 Two patients with the same amount of atrial fibrillation (AF) recurrence (same AF burden) but different recurrence patterns (low density AF [Patient B] vs. high density AF [Patient A]). The AF density as a metric can characterize and quantify the degree of temporal AF burden aggregation

$$\frac{e^{b_0+b_1\ln(days)}}{1+e^{b_0+b_1\ln(days)}}.$$

2.4 | Investigation effectors of AF episode length

To identify which factors may affect the length of the AF episodes, Poisson regression models were utilized. The following variables were investigated (gender, age, history of cardiac surgery (in the complete cohort of 1196 patients), presence of CAD, type of surgery (coronary artery bypass graft surgery, aortic valve surgery, mitral valve surgery, tricuspid valve surgery) and AF recurrence characteristics (AF burden, AF density) regarding their effect on AF episode length. The latter two factors (AF burden, AF density) were initially forced into all models to account for AF characteristics.

The *p* values of two-sided tests at a significance level of 0.05 are reported. Computer intensive bootstrapping was used to calculate confidence intervals. All statistical analyses were performed with R version 4.0.2 (R Development Core Team 2020).

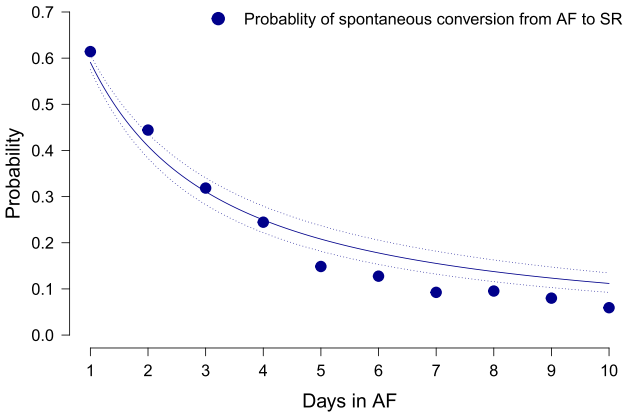


FIGURE 3 The more time a patient remains in AF the lower the probability that the patient will spontaneously convert to sinus rhythm without medical or interventional therapy. This probability reaches a plateau after about 5–7 days. Thereafter in our population the baseline (plateau) probability for spontaneous conversion remained stable. AF, atrial fibrillation; SR, sinus rhythm

3 | RESULTS

3.1 | Characteristics of AF recurrence

Three hundred and twenty patients were identified as having AF during a mean follow-up of 351 days (*SD*): 54 days; 409.6 patient × years). The mean AF burden in these patients was 0.21—the distribution of the AF burden is presented in Figure 2 (upper left). In general, high density AF was observed with a mean AF density of 0.78 (*SD*: 0.23), with its distribution shown in Figure 2 (right). Mean AF episode length was 5.9 days (interquartile range: 1), and was highly skewed with the vast majority of episodes being 1 day in duration (Figure 2, lower left).

3.2 | Probability of spontaneous conversion to SR and AF recurrence with time

For the whole patient population, the probability of spontaneous conversion from AF to SR (blue dots, Figure 3) significantly decreased the longer a patient remained in AF (Figure 3, dotted line, b_0: 0.34 95% confidence interval [CI]: 0.14 to 0.54; $p < 0.001$; b_1: −0.69 95% CI: −0.86 to −0.52; $p < 0.001$, in the patient cohort). Increased time spent in AF was associated with a lower likelihood of spontaneous conversion to sinus rhythm. The probability reached a plateau after about 5–7 days. Thereafter in our study population the baseline (plateau) probability for spontaneous conversion remained stable (Figure 3).

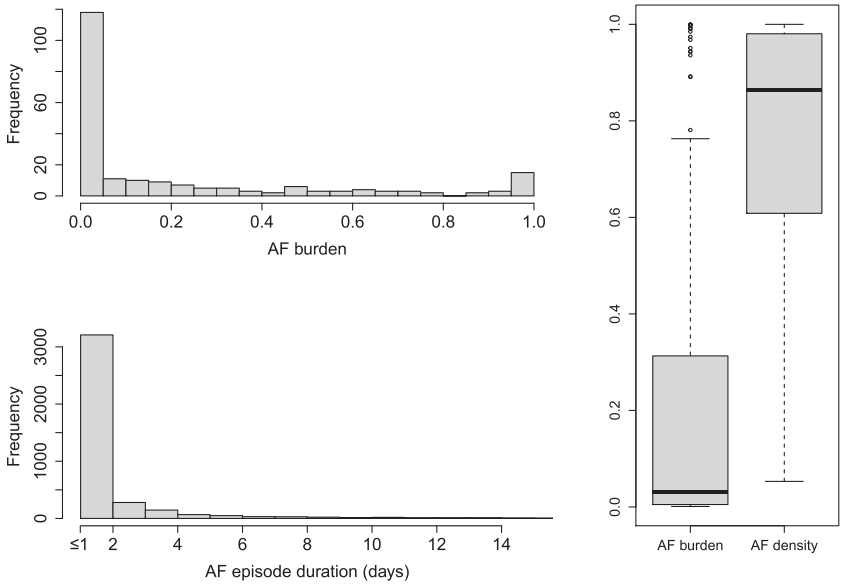


FIGURE 2 Distribution of AF burden (upper left), AF density (right) and AF episode duration (lower left) in the investigated patient population. AF, atrial fibrillation

TABLE 2 Poisson regression model of predictors of duration of time (days) spent in atrial fibrillation

Regressors	Estimate	95% Confidence interval	*p* value
History of coronary artery disease	0.83	0.72 – 0.95	0.006
History of myocardial infarction	1.15	1.06 – 1.26	0.001
Diabetes	1.11	1.02 – 1.21	0.022
Atrial fibrillation burden	11.82	10.27 – 13.59	<0.001
Aortic valve surgery	1.27	1.14 – 1.41	<0.001
Mitral valve surgery	1.13	1.00 – 1.28	0.043

3.3 | Factors affecting the length of AF episodes

The length of AF episodes was significantly influenced by AF characteristics and patient specific factors. An increased AF burden was associated with an increased AF episode duration (Table 2). Patient factors also influenced the length of AF episodes. History of myocardial infarction, diabetes as well as a history of aortic and mitral valve surgery were shown to be associated with AF episodes of longer duration—even after controlling for AF burden (Table 2). On the other hand, stable forms of CAD were associated with shorter episodes of AF (Table 2).

3.4 | Factors affecting the probability of spontaneous conversion to SR

Generally, we observed that the probability of spontaneous conversion to SR decreased the more time a patient spent in AF (Figure 3). AF characteristics (AF density, AF burden) as well as patient-related factors, such as age, presence of CAD, and LVEF significantly and independently affected the probability of spontaneous conversion to SR. The initial probability of spontaneous conversion was greater with higher AF density, presence of CAD, and increased age, and lower with increased burden. However, the higher the AF density, presence of CAD, advanced age and lower LVEF, the quicker the probability of spontaneous conversion to SR weans off as the patient spends more time in AF (Table 3). Figure 4 shows the probability of spontaneous conversion to SR with days spent in AF for two prototypical patients with the same AF characteristics (AF burden 0.2, AF density: 0.6) and age (70 years) but different clinical characteristics (Patient A: LVEF: 80% No CAD; Patient B: LVEF: 20% presence of CAD).

4 | DISCUSSION

To our knowledge, the present study is the only study evaluating AF recurrence patterns in patients with a history of cardiac surgery using data captured from continuous heart rhythm monitoring. More importantly, due to the fact that medical and interventional antiarrhythmic therapies were excluded from this patient population, this provided us with the unique opportunity to study the characteristics of AF recurrence, its natural history and factors that affect the probability of spontaneous conversion to SR. Continuous rhythm monitoring provides us the opportunity to quantify AF recurrence and AF temporal characteristics and evaluate patient and AF related factors that influence AF recurrence.

TABLE 3 Multivariable regression model of predictors of the probability of conversion from atrial fibrillation to sinus rhythm (=L/(1 + L) where $L = \exp(b_0 + b_1 \times \ln(\text{days}))$)

Regressors	Estimate	95% Confidence interval	*p* value
Dependent variable: b0			
Atrial fibrillation burden	−2.2	−2.5 to −1.97	<0.001
Atrial fibrillation density	1.15	1.08 to 1.28	<0.001
Coronary artery disease	0.18	0.07 to 0.29	<0.001
Patient age	0.007	0.002 to 0.012	<0.008
Dependent variable: b1			
Atrial fibrillation density	−1.2	−1.48 to −1.08	<0.001
Coronary artery disease	−0.11	−0.22 to −0.01	0.02
Patient age	−0.007	−0.012 to −0.001	0.02
Left ventricular ejection fraction	0.005	0.002 to 0.008	<0.001

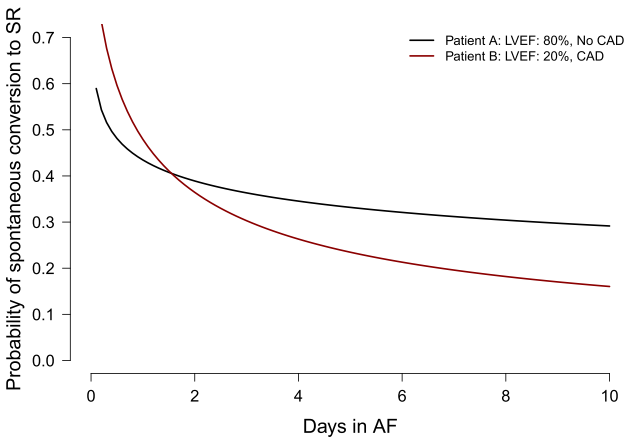


FIGURE 4 Atrial fibrillation characteristics and patient related factors significantly and independently affect the probability of spontaneous conversion to SR. The figure presents a graphical display of the models presented in Table 3. Two prototypical patients with the same AF characteristics (AF burden 0.2, AF density: 0.6) and age (70 years) but different clinical characteristics (Patient A: LVEF: 80% no CAD; Patient B: LVEF: 20% presence of CAD) result in a different probability of spontaneous conversion to SR with time spent in AF. AF, atrial fibrillation; CAD, coronary artery disease; LVEF, left ventricular ejection fraction; SR, sinus rhythm

4.1 | Influence of patient characteristics and history of cardiac surgery

The analysis of AF episode duration (Table 2) revealed that even for the same amount of AF burden, patient-related and AF-related factors influence the duration of AF episodes and probabilities of AF termination. In the present study, well known factors associated with AF,[4] such as CAD, diabetes and myocardial infarction could be validated. In our analyses, diabetes and history of myocardial infarction were found to be associated with longer episodes of AF, whereas stable forms of CAD were associated with shorter AF episodes.

While we could not observe a difference in AF burden between patients with or without mitral or aortic valve interventions, these patients had significantly longer AF episodes than patients without valve interventions (Table 2). This finding provides a different perspective to the well-known fact that valve disorders are risk factors for the development of AF[4] and influence the pathophysiology of AF recurrence and its natural history. More importantly this effect was independent from the observed AF burden, a fact that might suggest that valve surgery and significant valvular heart disease in general may affect not only AF recurrence but also AF characteristics and recurrence patterns.

4.2 | AF, patient characteristics and the risk of stroke

While our study as well as the initial OMNI and TRENDS studies could not reveal a direct association between stroke and amount of AF,[11,12] significant evidence has recently emerged to suggest that stroke risk is a function of time spent in AF as well as the patient-related factors captured in the CHA_2DS_2-VASc score.[13] In non-anticoagulated patients, the stroke risk increases with more time spent in AF as well as with higher CHA_2DS_2-VASc scores. Higher maximum daily AF burden (which would correspond to a higher AF density) has been associated with higher stroke event rates,[13] suggesting that patients with intermediate CHA_2DS_2-VASc scores may benefit from continuous heart rhythm monitoring to further refine and define their stroke risk. Currently cardiac surgery is facing a patient population shift, with older patients increasingly receiving interventional therapies and younger patients typically undergoing surgical procedures. In the young patients receiving surgical treatment, the finding of high density postoperative AF recurrence may signal a higher risk of stroke if the patients receive no anticoagulation.[13] The understanding that high density AF may have an influence on stroke risk may be important for young patients with low CHA_2DS_2-VASc scores receiving valve surgery who may typically not receive anticoagulation. This concept and management strategies in this respect require intense further investigation.

4.3 | The importance of temporal AF characteristics

In the patient population of the present study, 75% of patients experienced AF and subsequent AF recurrence. While most episodes terminated within a day, the majority of AF manifested as high density AF (Figure 1 and Figure 2). We have previously shown that intermittent rhythm monitoring fails to detect AF in a large proportion of patients, and AF detection rates were significantly lower in patients with high density AF.[5] Currently there are no recommendations regarding postoperative rhythm monitoring strategies in cardiac surgery patients who do or do not have a history of AF. Patients with AF and a high CHA_2DS_2-VASc score are likely to receive anticoagulation. Based on the current work and our previous work, patients with a low CHA_2DS_2-VASc score and a history of AF may benefit from postoperative continuous rhythm monitoring to quantify AF and evaluate whether anticoagulation is beneficial. More data is required to develop such strategies and underpin their value in the clinical setting.

4.4 | A framework to evaluate success of medical and interventional therapies for AF

Most studies investigating AF recurrence or the success of medical or interventional therapies for AF consider a binary outcome (AF recurrence: yes/no). Continuous monitoring devices provide a wealth of data far superior to the binary AF concept. Using the data from continuous monitoring devices, a quantitative approach to both AF burden and AF temporal aggregation can be investigated and AF characteristics and influence of medical or interventional therapies can be further characterized in terms of these two quantitative metrics. Additionally, under the prism of the insights from our study, using data from CM devices inferences on AF episode length and probability of conversion to SR can also be thoroughly investigated in patients receiving medical or interventional AF therapies. Data from CM devices provide vastly richer data to investigate hypotheses regarding AF recurrence and treatments in a research environment and this data has a strong potential to affect clinical practice. However, the benefit of this approach remains to be seen in future studies.

4.5 | The need for a better understanding of AF recurrence in patients after cardiac surgery

New-onset postoperative AF can be found in approximately 33% of the patients after cardiac surgery.[14] Any form of cardiac surgery may lead to atrial electrical instability which must be seen as the precursor to the rhythm disorder which we then detect as AF. The cause for this postoperative instability is not yet completely understood. Current data regarding the incidence and

characteristics of postoperative AF stem from Holter or telemetry studies and as mentioned above are thusly not reliable enough for the investigation of this issue. It is critical that future studies apply continuous monitoring for the investigation of AF patterns. In this mindset we have initiated a study of new-onset AF after coronary artery bypass graft surgery via continuous rhythm monitoring using insertable cardiac monitors (CABG-AF, DRKS00018887). It is only with such studies, performing continuous rhythm monitoring, that high quality data can be collected and the true nature of AF in different population subsets can be understood. We believe that with the data from the cardiac surgery population subset of the OMNI and TRENDS trials presented above we could add to the general knowledge regarding AF recurrence. We hope that through the evaluation of further continuous monitoring data in the future our gap in knowledge regarding postoperative arrhythmia can be reduced.

5 | CONCLUSION

Short high density AF episodes are common in patients with a history of cardiac surgery who have developed AF. Spontaneous conversion to SR diminishes with increasing time spent in AF and is influenced by both AF characteristics and several patient-related factors. Aortic and mitral valve surgery are associated with longer AF episodes in this patient cohort. CM can provide detailed information about AF recurrence that may influence clinical management. Investigation of cardiac surgery patients using CM in a controlled research setting has the potential to facilitate a better understanding of AF. It is crucial that we apply continuous monitoring strategies in future investigations of AF in all population subsets.

6 | LIMITATIONS

The time point of cardiac surgery was not captured in the database of the OMNI and TRENDS trials. The only available time-related information is that PM/ICD implantation was performed at least 3 months after cardiac surgery i.e. continuous monitoring began at least 3 months after surgery. A further central limitation is the fact that the presence of a PM/ICD may have an influence on the development and persistence of AF. No data was available regarding pacing burden and type of pacing i.e. atrial or ventricular (both would be of great interest in this respect). Due to data acquisition limitations of the devices used in the study, a day (24 h) was the finest temporal resolution available for all analyses.

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CONFLICT OF INTERESTS

Paul D. Ziegler is a full time employee and shareholder of Medtronic (>15.000 USD). Efstratios I. Charitos, MD, PhD is a shareholder of Medtronic (>15.000 USD).

DATA AVAILABILITY STATEMENT

The present work is an analysis of data collected in the OMNI and TRENDS multicenter trials. Due to the retrospective nature of this analysis, data and study materials cannot be made available to other researchers for purposes of reproducing the results or replicating the procedure.

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